

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

- | | |
|-------------------------------------|--|
| n/a | Confirmed |
| ☐ | ☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement |
| ☐ | ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly |
| ☐ | ☒ The statistical test(s) used AND whether they are one- or two-sided
<i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i> |
| ☐ | ☒ A description of all covariates tested |
| ☐ | ☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons |
| ☐ | ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) |
| ☐ | ☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted
<i>Give P values as exact values whenever suitable.</i> |
| ☒ | ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☐ | ☒ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated |

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection

We recruited 121,556 pregnancy participants from Shenzhen Baoan Women’s and Children’s Hospital (referred to as Baoan) and Longgang District Maternity and Child Healthcare Hospital of Shenzhen City (referred to as Longgang), who underwent NIPT test (0.1x sequencing depth) during the first or second trimester (mean: 12 weeks of gestation, 95% CI 11-13 weeks) between 2017 and 2021. After excluding 5,412 participants with multiple NIPT sequencing records, 116,144 women were included in our study.

Data analysis

The code and software used in this study are publicly available, and the URLs are listed below:
 Methods for analyzing the NIPT data for GWAS: <https://github.com/liusylab/NIPT-human-genetics>
 GDM prediction model: https://github.com/liusylab/GDM_risk_prediction.
 PLINK2: <https://www.cog-genomics.org/plink/2.0/>
 GLIMPSE (version 1.1.1): <https://odelaneau.github.io/GLIMPSE/glimpse1/>
 GCTA: <https://yanglab.westlake.edu.cn/software/gcta/#COJO>
 METAL: <https://github.com/statgen/METAL>
 LDlink: <https://ldlink.nci.nih.gov/>
 The GWAS Catalog: <https://www.ebi.ac.uk/gwas/>
 locuszoom: <https://genome.sph.umich.edu/wiki/LocusZoom>
 linemodels: <https://github.com/mjpirinen/linemodels>
 colocalization analysis: <https://cran.r-project.org/web/packages/coloc/index.html>
 Metascape website: <https://metascape.org/>
 PRSice-2: <https://choishingwan.github.io/PRSice/>

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

The complete GWAS summary statistics for GDM, FPG, OGTT0H, OGTT1H, OGTT2H, and HbA1c have been deposited in the GWAS Catalog database (<https://www.ebi.ac.uk/gwas/>) with accession numbers: GCST90566419[<https://www.ebi.ac.uk/gwas/studies/GCST90566419>](GDM), GCST90566420[<https://www.ebi.ac.uk/gwas/studies/GCST90566420>](FPG), GCST90566421[<https://www.ebi.ac.uk/gwas/studies/GCST90566421>](OGTT0H), GCST90566422[<https://www.ebi.ac.uk/gwas/studies/GCST90566422>](OGTT1H), GCST90566423[<https://www.ebi.ac.uk/gwas/studies/GCST90566423>](OGTT2H), GCST90566424[<https://www.ebi.ac.uk/gwas/studies/GCST90566424>](HbA1c), with approval from the China's National Health Commission (permission number: 2025BAT00034-BG1).

Raw sequencing data have been deposited in the Genome Sequence Archive (GSA) for Humans at the National Genomics Data Center under the BioProject accession number GSA-Human: HRA006833[<https://ngdc.cncb.ac.cn/gsa-human/browse/HRA006833>](Non-invasive prenatal testing sequencing data), with approval from China's National Health Commission (permission number: 2024BAT01079). This study complies with the Principle for the Access of Human Genetic Resource Data in NGDC(https://ngdc.cncb.ac.cn/gsa-human/document/Principle_of_Accessing_Human_Genetic_Resource_Data_in_NGDC_V1.pdf). The dataset is restricted to research use by the designated research group and its collaborators. Redistribution or use for individual identification is prohibited. Access to the data requires a formal application following GSA guidelines (https://ngdc.cncb.ac.cn/gsa-human/document/GSA-Human_Request_Guide_for_Users_us.pdf). Applications are reviewed by both the system and the Data Access Committee (DAC). Upon approval, data will be made available for download. Data access is granted for academic research purpose only. If no response is received from the DAC within two weeks, the system will issue an automatic reminder. The data download window is valid for three months; users must complete the download within this period, otherwise reapplication will be required. Full procedures and condition are detailed in the GSA guidelines.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender

All participants included in the study were Chinese pregnant women.

Reporting on race, ethnicity, or other socially relevant groupings

All participants included in the study were Chinese.

Population characteristics

Most participants were between the ages of 25-34, and the pregnant women were underwent NIPT test (0.1x sequencing depth) during the first or second trimester (mean: 12 weeks of gestation, 95% CI 11-13 weeks) between 2017 and 2021.

Recruitment

We recruited 121,556 pregnancy participants from Shenzhen Baoan Women's and Children's Hospital (referred to as Baoan) and Longgang District Maternity and Child Healthcare Hospital of Shenzhen City (referred to as Longgang). These participants underwent NIPT with an average of 0.17x sequencing depth during the first or second trimester (mean gestational age: 16 weeks) between 2017 and 2021. After excluding 5,412 participants with multiple NIPT sequencing records, 116,144 women remained in the study (Fig. 2). The Longgang cohort included 7,438 GDM cases (15.5%) and 40,446 controls, while the Baoan cohort included 4,586 GDM cases (14.4%) and 27,399 controls. Demographic data and measurements for 43 biomarkers collected before 20 weeks of gestation were also obtained (Supplementary Data 1).

Ethics oversight

This study was approved by the Ethics Committee of the School of Public Health (Shenzhen), Sun Yat-Sen University (approval number: 2021. No.8), as well as the Institutional Review Boards of Shenzhen Baoan Women's and Children's Hospital (approval number: LLSC2021-04-01-10-KS) and Longgang District Maternity and Child Healthcare Hospital (approval number: LGFYXLLL-2022-024).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

- ☒ Life sciences
- ☐ Behavioural & social sciences
- ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	We recruited 121,556 pregnancy participants from Shenzhen Baoan Women’s and Children’s Hospital (referred to as Baoan) and Longgang District Maternity and Child Healthcare Hospital of Shenzhen City (referred to as Longgang). These participants underwent NIPT with an average of 0.17x sequencing depth during the first or second trimester (mean gestational age: 16 weeks) between 2017 and 2021. After excluding 5,412 participants with multiple NIPT sequencing records, 116,144 women remained in the study (Fig. 2). The Longgang cohort included 7,438 GDM cases (15.5%) and 40,446 controls, while the Baoan cohort included 4,586 GDM cases (14.4%) and 27,399 controls. Demographic data and measurements for 43 biomarkers collected before 20 weeks of gestation were also obtained (Supplementary Data 1). Furthermore, we collected data from two independent cohort. The first cohort, referred to as the Baoan 20K cohort, included 20,439 participants recruited after 2021 from Shenzhen Baoan Women’s and Children’s Hospital and was not included in the Baoan cohort. After excluding participants without a clear GDM diagnosis or those who underwent NIPT after 20 weeks of gestation, 14,129 participants remained for training and evaluating the prediction model. The second cohort, referred to as the NIPT PLUS cohort, served as an external validation cohort. This cohort included 5,897 individuals who underwent deeper sequencing (0.3x), comprising 795 GDM cases (22.3%) and 2,770 controls with clear GDM diagnosis records before 20 weeks of gestation.
Data exclusions	Excluding 5,412 participants with multiple NIPT sequencing records.
Replication	To evaluate and replicate the GWAS meta-analysis results for GDM and five glycemic traits, we first compared the genetic effect estimates of the lead SNPs between the separate GWAS conducted at the two hospitals and the meta-analysis results. For external replication, we performed replication analyses in two independent cohorts (Baoan 20K and NIPT PLUS) and compared the meta-analysis results of these cohorts. Additionally, we compared the GWAS meta-analysis results for FPG and HbA1c with summary statistics from the Taiwan Biobank, and for OGTT2H with summary statistics from the MAGIC consortium’s East Asian population. The GDM GWAS findings were further replicated using data from a previous European study.
Randomization	This is not relevant since we did not use different experimental groups or conditions in our study.
Blinding	This is not relevant since we did not use different experimental groups or conditions in our study.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems	Methods
n/a	Involvement in the study
☒ ☐ Antibodies	☒ ☐ ChIP-seq
☒ ☐ Eukaryotic cell lines	☒ ☐ Flow cytometry
☒ ☐ Palaeontology and archaeology	☒ ☐ MRI-based neuroimaging
☒ ☐ Animals and other organisms	
☒ ☐ Clinical data	
☒ ☐ Dual use research of concern	
☒ ☐ Plants	

Plants

Seed stocks

Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.

Novel plant genotypes

Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.

Authentication

Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.